
Spatial Analysis of the Network Efficiency of the Left-Bank Bicycle Path along the Dahan River

1. INTRODUCTION

With the growing awareness of sustainable transportation and recreational activities, bicycle paths have become an important part of urban green transportation systems and public spaces. The left-bank bicycle path along the Dahan River is one of the major riverside cycling corridors in New Taipei City, extending through Xinzhuang, Shulin, and Yingge Districts. Different sections of the path serve different functions: the Xinzhuang section is mainly commuter-oriented, while the Shulin and Yingge sections are primarily used for leisure and recreation.

However, despite the development of the cycling network, its convenience may still be affected by geographic conditions and spatial configurations. Factors such as the uneven distribution of levee-crossing facilities and indirect access routes may increase the time and effort required for users to access and use the bicycle path, thereby affecting its accessibility as both a transportation and recreational facility.

Therefore, this study analyzes the Dahan River left-bank bicycle path from a spatial perspective. Using Geographic Information System (GIS) network analysis, this research evaluates the spatial accessibility and network efficiency of the bicycle path, explores differences in access costs among different areas, and identifies inefficient sections and potential improvement points within the network as references for future planning.

2. Methodology

This study focuses on the left-bank bicycle path along the Dahan River. Bicycle path data and the study area layer were integrated to construct a network model for spatial analysis. Since the data are primarily spatial in nature, distance was used as the main basis for analysis.

The research consists of three parts: accessibility analysis of bicycle path entrances, network efficiency analysis, and isochrone analysis. In addition, the distribution of YouBike stations was included as a reference for evaluating the integration between the public bicycle system and the bicycle path network. The results were then interpreted comprehensively to assess the overall performance of the network.

(1) Entrance Accessibility Analysis

Representative transportation nodes within the study area, such as MRT stations and railway stations, were selected as starting points, while the nearest bicycle path entrances (e.g., levee crossings and

underpass connections) were designated as destinations. A shortest path analysis was conducted to calculate the shortest distance from each transportation node to the nearest bicycle path entrance. By comparing the average distances among different areas, the spatial accessibility of the bicycle path network can be evaluated. Longer distances indicate higher access costs.

(2) Network Efficiency Analysis

Bicycle path entrances were treated as network nodes, and adjacent nodes were paired for analysis. The shortest route distance along the network and the straight-line distance between nodes were calculated and evaluated using the Detour Index:

$$\text{Detour Index} = \text{Network Distance} \div \text{Straight-Line Distance}$$

A higher Detour Index indicates a more indirect route and lower network efficiency. Comparisons among different sections help identify areas with poor connectivity or excessive detours.

(3) Service Area Analysis

To evaluate the service coverage of bicycle path entrances along the left bank of the Dahan River, service areas were established around each entrance.

Considering that the average cycling speed for recreational cyclists is approximately 15 km/h, a 10-minute cycling time was converted into an accessible distance of approximately 2,500 meters. GIS buffer analysis was then applied to create service areas with a radius of 2,500 meters around each entrance, representing the approximate area that can be reached within 10 minutes.

The resulting service areas were used to visualize the spatial coverage of bicycle path entrances and to examine differences in service density among administrative districts. This analysis provides a basis for evaluating whether the entrance distribution is balanced and for identifying potentially underserved areas.

3. DATA COLLECTION

The spatial data used in this study include the bicycle path network, entrance nodes, YouBike stations, and representative transportation nodes. These datasets were integrated and analyzed using open-data platforms and Geographic Information Systems (GIS). The data sources and platforms are described as follows.

(1) Bicycle Path Network Data

The bicycle path network data were primarily obtained from OpenStreetMap (OSM) and the New Taipei City Open Data Platform. OSM provides comprehensive and editable road and bicycle path layers, serving as the foundation for network analysis. Bicycle-related spatial data provided by the New Taipei City Government were also used to improve the completeness and spatial accuracy of the riverside bicycle path network.

(2) Bicycle Path Entrance Nodes (Destinations)

The locations of bicycle path entrances, including levee crossings, underpass connections, and connecting roads, were obtained from OpenStreetMap (OSM) and bicycle-related datasets provided by the New Taipei City Open Data Platform. Satellite imagery and GIS basemaps were further used to identify and verify entrance locations and to establish the node dataset required for network analysis.

(3) YouBike Station Data

YouBike station data were mainly collected from OSM transportation node layers (bicycle rental / YouBike stations). Public map services, such as Google Maps and the official YouBike station map, were used for location verification and correction. These stations represent important access points to the public bicycle-sharing system and reflect actual cycling origins. By integrating entrance and YouBike station data, the spatial connectivity between the public bicycle system and the riverside bicycle path network can be further examined.

(4) Representative Nodes (Origins)

For the entrance accessibility analysis, transportation hubs within each district were selected as representative origin points, including MRT stations and railway stations. Their location data were obtained from:

- OSM transportation node data (MRT stations and railway stations)
- Public mapping platforms, such as Google Maps, for location verification and correction

These transportation nodes represent major passenger gathering points and serve as reasonable origins for accessing the bicycle path network.

(5) Study Area Data

The study area focuses on the left bank of the Dahan River, covering the districts of Xinzhuang, Shulin, and Yingge. River boundary data and basemaps were obtained from OSM and QGIS basemap services. The datasets were clipped and processed to define the spatial extent of the study area.

(6) Data Processing and Analysis Platform

QGIS was used as the primary GIS platform for data processing and spatial analysis. The main analytical procedures included:

- Spatial data integration and layer processing
- Network construction and shortest path analysis
- Entrance accessibility analysis
- Network efficiency analysis using the Detour Index
- Service area and isochrone analysis

Spreadsheet software, such as Microsoft Excel, was also used for distance calculations and result organization.

4. Results and Discussion

(1) Entrance Accessibility Analysis

大漢溪左岸自行車道交通節點至出入口最短路徑分析圖



Figure 1.

Figure 1 presents the results of the shortest path analysis from MRT stations and railway stations within the study area to the entrances of the Dahan River left-bank bicycle path. Table 1 summarizes the shortest distances between each transportation node and its corresponding bicycle path entrance. Overall, significant differences in accessibility were observed among transportation nodes, indicating that the degree of connectivity between transportation facilities and the bicycle path network varies across different areas.

Station	District	Distance (m)
捷運頭前庄	新莊區	231
捷運鶯歌	鶯歌區	274
捷運新莊	新莊區	407
捷運先嗇宮	三重區	575
捷運陶瓷老街	鶯歌區	588
台鐵鶯歌	鶯歌區	776
台鐵山佳	樹林區	1250

台鐵南樹林	樹林區	1302
台鐵樹林	樹林區	1304
捷運輔大	新莊區	2356
捷運丹鳳	新莊區	2517
捷運迴龍	新莊區	3897
台鐵鳳鳴	鶯歌區	3972

Table 1.

As shown in Table 1, the transportation nodes with the shortest distances to bicycle path entrances were Touqianzhuang Station (231 m), Yingge Station (274 m), and Xinzhuang Station (407 m). These stations provide convenient access to the bicycle path network, indicating a relatively strong level of connectivity. Touqianzhuang Station and Xinzhuang Station are located near the Dahan River waterfront area, resulting in lower access costs. Although Yingge Station is located at the southern end of the study area, its bicycle path entrance is well connected to the surrounding road network, contributing to its high accessibility.

In contrast, several transportation nodes were located much farther from bicycle path entrances, including Fengming Station (3,972 m), Huilong Station (3,897 m), and Danfeng Station (2,517 m). Although these stations are situated within or near the study area, they lack direct connections to the bicycle path network. Users must travel longer distances through the road system before reaching a bicycle path entrance, resulting in higher access costs. This finding suggests that the integration between bicycle path entrances and public transportation facilities could be further improved.

When examined by district, the stations in Shulin District showed relatively consistent distances, ranging from approximately 1,250 m to 1,304 m, indicating a more balanced distribution of bicycle path entrance services. In contrast, Xinzhuang District exhibited considerable variation in accessibility. While Touqianzhuang Station and Xinzhuang Station were both located within 500 m of a bicycle path entrance, Danfeng Station and Huilong Station were located 2,517 m and 3,897 m away, respectively. This highlights significant spatial differences in accessibility within the district. Yingge District displayed a similar pattern of variation. Yingge Station and Ceramic Old Street Station were located only 274 m and 588 m from the nearest bicycle path entrances, whereas Fengming Station was located as far as 3,972 m away. This suggests that bicycle path entrance services are primarily concentrated around existing urban and tourist areas.

Overall, the results indicate that the Dahan River left-bank bicycle path has established a basic level of connectivity with public transportation nodes. However, substantial differences in accessibility remain among different areas. Future improvements, such as adding new entrances or enhancing connecting road networks in areas with relatively poor accessibility, including Huilong and Fengming, could strengthen the integration between the bicycle path network and public transportation systems and further improve user convenience.

(2) Network Efficiency Analysis

大漢溪左岸自行車道 路網效率分析圖

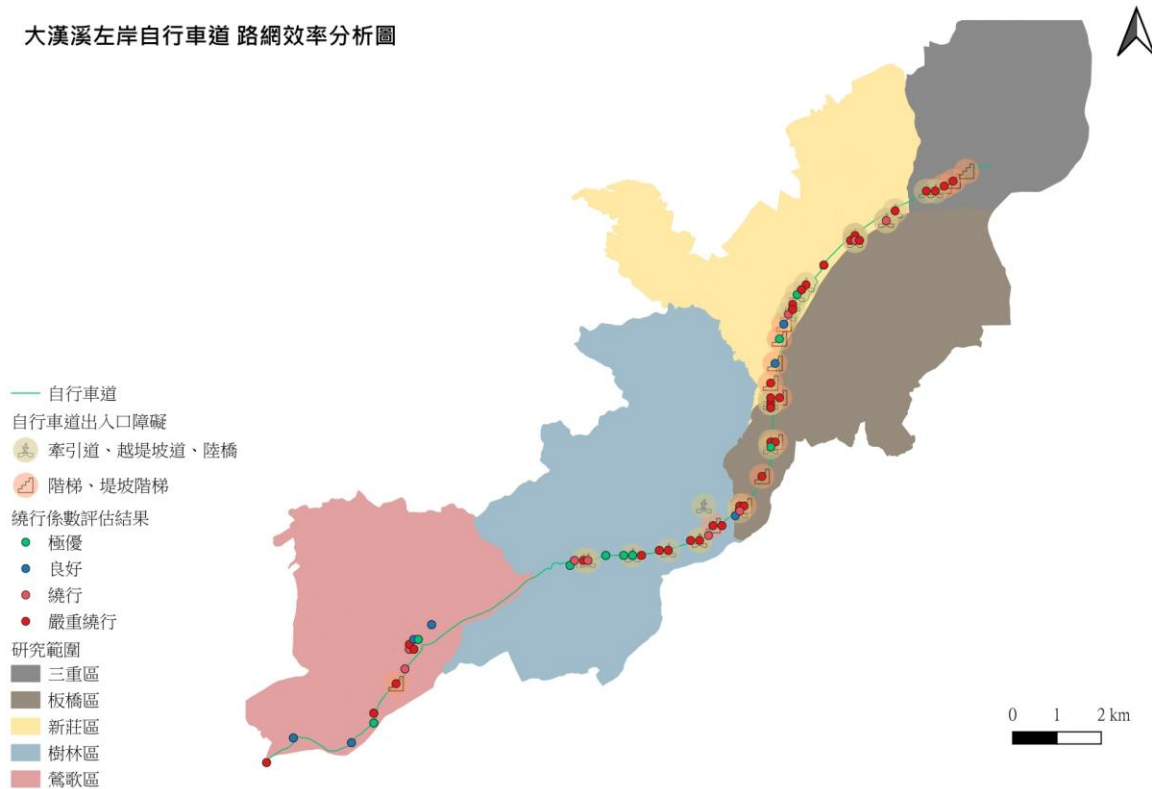


Figure 2.

According to the results shown in Figure 2, network performance is closely related to the type of bicycle path entrance facilities. The evaluation criteria for the Detour Ratio used in this study were based on previous studies of non-motorized transportation and urban network geometry. Research on urban network circuitry has suggested that mature urban transportation networks typically have average detour ratios ranging from 1.20 to 1.30 (Newell, 1980; O'Sullivan & Morrall, 1996). Therefore, detour ratios between 1.11 and 1.30 were classified as representing a good level of network efficiency. When the detour ratio exceeds 1.30, users are more likely to perceive routes as excessively indirect and inconvenient (Chou et al., 2023).

The network analysis results indicate that sections with excellent or good efficiency are generally located near at-grade entrances or well-designed levee-crossing bridges, which provide direct and convenient access to the bicycle path system. In contrast, low-efficiency bottleneck areas are commonly associated with entrances that consist only of stairways or levee-slope stairs. Due to the lack of barrier-free ramps or bicycle-access facilities, combined with elevation differences and physical barriers created by flood-control infrastructure, cyclists are often required to make substantial detours, reducing overall network efficiency and convenience.

A closer examination of local areas reveals that the boundary region between Sanchong, Xinzhuang, and Banqiao Districts contains a high concentration of stairway-type entrances. As shown in Figure 2, these locations correspond to clusters of high detour-ratio nodes classified as severe detours (>1.50) and moderate detours (1.31–1.49). This pattern suggests that levees and other flood-control structures act as major physical barriers between urban areas and riverside green spaces. As a result, cyclists experience difficulty maintaining

direct and continuous routes, particularly for short-distance trips and daily commuting, thereby reducing overall network performance.

In comparison, the central area of Shulin District and the southern section of Yingge District exhibit more continuous distributions of low detour-ratio nodes. These areas are generally associated with facilities such as bicycle ramps, levee-crossing slopes, and pedestrian bridges, indicating that modern and cyclist-friendly access infrastructure has been incorporated into the network. Consequently, route continuity and travel efficiency in these areas are noticeably better than those observed in the upstream sections.

For the key nodes identified as severe detour locations, it is recommended that planners prioritize the construction of bicycle ramps and barrier-free access facilities to reduce spatial barriers. In addition, improved wayfinding systems and directional signage should be provided to help cyclists identify optimal routes more efficiently. Through the enhancement of entrance facilities and navigation systems, the overall accessibility, connectivity, and user experience of the Dahan River left-bank bicycle path network can be significantly improved.

Evaluation Result	Detour Ratio Range	Spatial Characteristics	Improvement Priority
Excellent	1.00–1.10	Direct access through barrier-free ramps and levee-crossing bridges	Low
Good	1.11–1.30	Basic access facilities available; detour distance remains acceptable	Moderate
Detour Required	1.31–1.49	Affected by levees and stairways; users must detour to nearby crossing facilities	Medium–High
Severe Detour	> 1.50	Strongly affected by levees and stairways; users must make substantial detours to reach crossing facilities	High

(3) Service Area Analysis

大漢溪左岸自行車道出入口10分鐘服務範圍分析圖

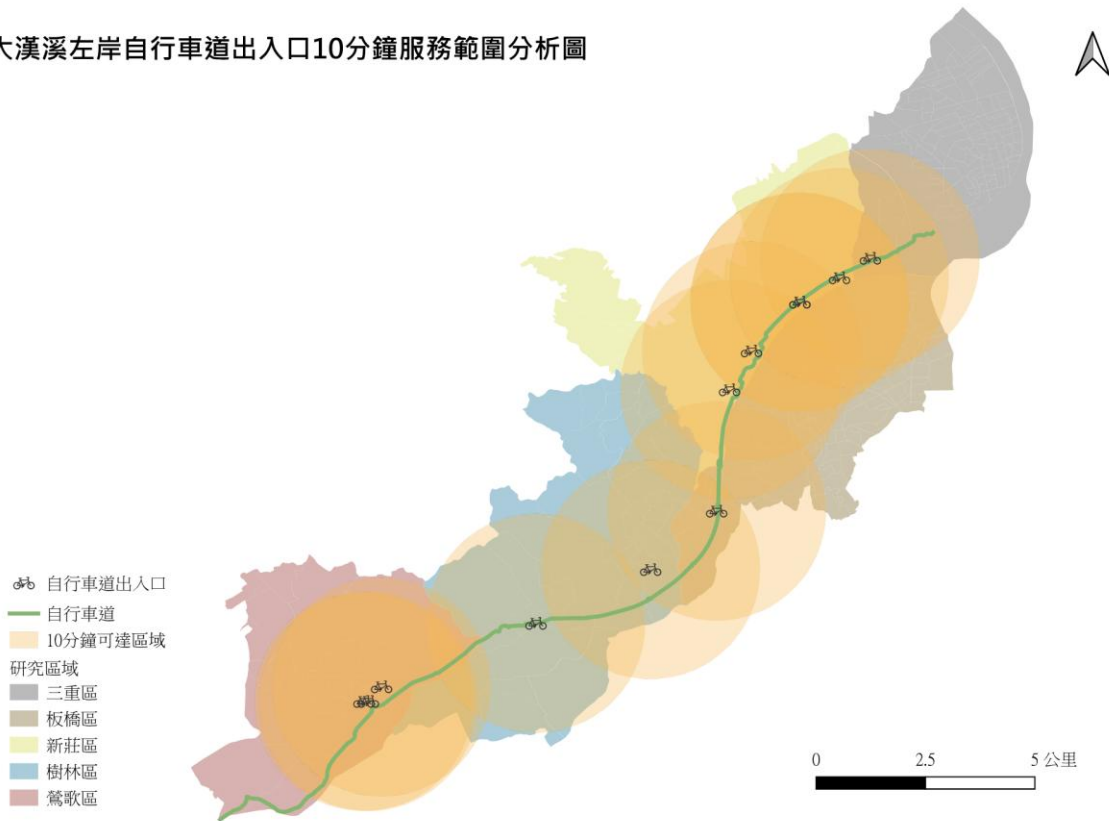


Figure 3.

Figure 3 presents the results of the 10-minute service area analysis for bicycle path entrances associated with the selected transportation nodes. Based on an average recreational cycling speed of 15 km/h, a 10-minute cycling distance was converted into an accessible distance of approximately 2,500 meters. Previous studies have suggested that 10 minutes represents an acceptable travel-time threshold for using bicycles as a feeder mode to public transportation, effectively extending the service radius of transit stations to approximately 2.5 km (Martens, 2004). GIS buffer analysis was therefore applied to establish service areas around each bicycle path entrance and to evaluate the spatial coverage of the bicycle path network.

As shown in Figure 3, the service areas are distributed continuously along the left-bank bicycle path of the Dahan River, with noticeable overlap between adjacent entrances. This pattern indicates a relatively high level of spatial continuity between transportation nodes and bicycle path entrances. As a result, users can conveniently access the bicycle path network from MRT stations and railway stations, contributing to the development of an integrated green transportation system.

A closer examination of the service area distribution reveals substantial overlap in the northern sections of Banqiao, Xinzhuang, and Shulin Districts. This suggests that bicycle path entrances are more densely distributed in these areas and that connections between transportation nodes and the bicycle path network are relatively strong. In contrast, several sections near the southern end of the study area in Yingge District fall outside the service areas defined in this study. This does not necessarily indicate a lack of bicycle path entrances, but rather reflects weaker connections between these areas and the representative transportation nodes selected for analysis. Compared with the northern sections, the southern portions of the bicycle path

network are less directly connected to MRT stations and railway stations, resulting in a lower degree of transportation integration.

Overall, the results indicate that the selected transportation nodes and their corresponding bicycle path entrances provide coverage for most sections of the Dahan River left-bank bicycle path. This suggests that the bicycle path network has achieved a certain level of integration with the surrounding public transportation system. However, regional differences remain evident. The northern sections of Banqiao, Xinzhuang, and Shulin benefit from denser transportation nodes and therefore exhibit better service coverage and transfer opportunities. In contrast, some sections of Yingge District remain less connected to major transportation hubs. Future improvements in transfer facilities and stronger integration between the bicycle network and public transportation system, particularly in the southern section of the study area, could further enhance accessibility and service quality.

5. Conclusion

By integrating the results of the entrance accessibility analysis, network efficiency analysis, and service area analysis, together with the spatial distribution of YouBike stations, this study provides a comprehensive evaluation of the integration between the Dahan River left-bank bicycle path and the surrounding transportation system. The results indicate that the bicycle path network possesses a high degree of continuity, with most sections effectively connected through multiple entrances to form a complete cycling network. Although some entrances are still affected by facilities such as stairways, pedestrian bridges, and levee-crossing structures that reduce user convenience, the overall system demonstrates satisfactory accessibility and network efficiency. Furthermore, the bicycle path has achieved a certain level of integration with public transportation nodes and the YouBike system, forming an interconnected green transportation network.

Key Findings

(1) Regional Differences in Spatial Configuration and Service Quality

The integration between the bicycle path network and public transportation facilities exhibits a clear spatial pattern characterized by stronger connectivity in the northern section and weaker connectivity in the southern section. Xinzhuang, Banqiao, and northern Shulin benefit from a higher concentration of transportation nodes, greater accessibility, and substantial overlap among 10-minute service areas (2,500 m). For example, Touqianzhuang Station and Xinzhuang Station are both located within 500 m of bicycle path entrances. These characteristics contribute to a well-developed green transportation network.

In addition, most YouBike stations are concentrated in densely populated areas with a high number of transportation nodes. This suggests that the bicycle-sharing system has become an important feeder mode connecting public transportation and the bicycle path network, improving the convenience of accessing the cycling system. In contrast, Yingge and several peripheral areas, including Huilong Station, Fengming Station, and Danfeng Station, are located much farther from bicycle path entrances, often requiring travel distances

approaching or exceeding 4 km. The lack of direct connections results in lower service coverage and reduced transfer convenience, limiting the level of transportation integration in these areas.

(2) Entrance Design as a Key Factor Affecting Network Efficiency

The results indicate a strong relationship between network efficiency and the physical design of bicycle path entrances. Sections equipped with at-grade entrances, bicycle ramps, or levee-crossing bridges generally exhibit excellent or good network efficiency, providing cyclists with direct and convenient access routes. In contrast, entrances consisting of stairways, pedestrian bridges, levee-slope stairs, or other elevated crossing facilities often create significant detours and become bottlenecks within the network. This issue is particularly evident near the boundary between Sanchong and Banqiao Districts.

When combined with the accessibility analysis, the findings reveal that many of these network-efficiency constraints occur in the middle and upper sections of the Dahan River corridor, which also contain relatively dense populations and transportation facilities. Therefore, improving entrance design and reducing detour requirements in these areas should be considered a priority for future bicycle network planning and development.

大漢溪左岸YouBike站點分布圖

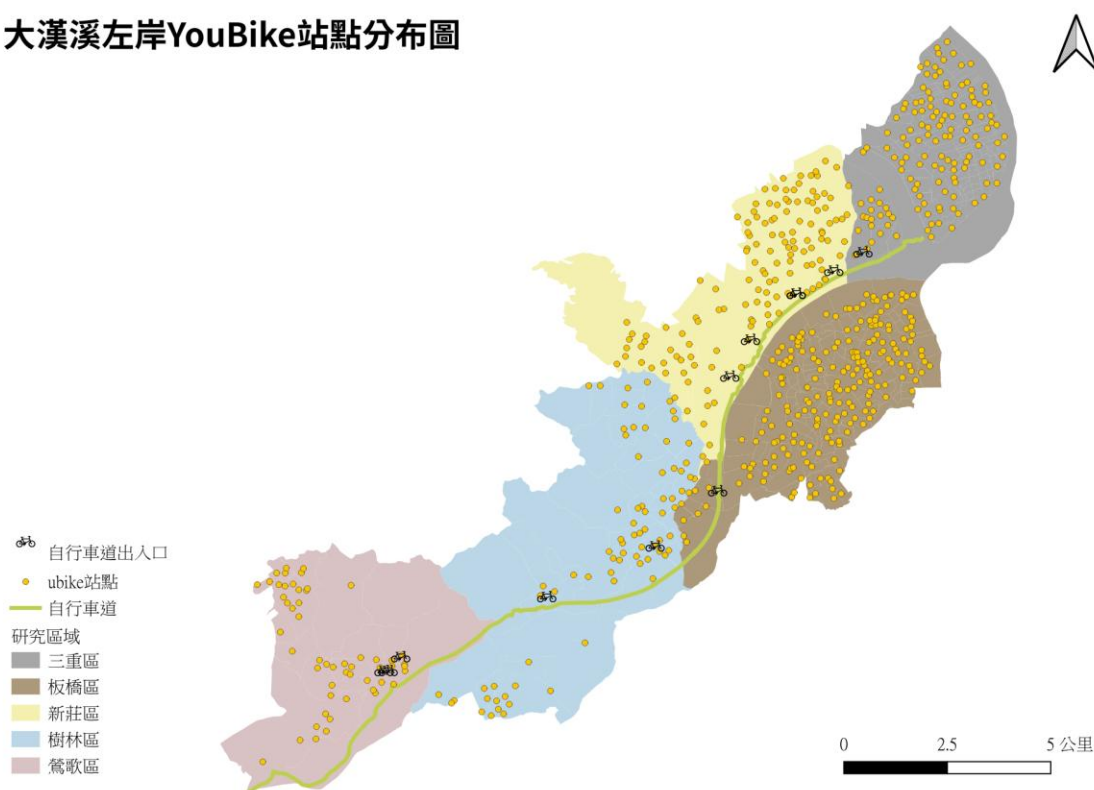


Figure 4.

Future Improvement Strategies

To enhance the overall cycling environment, stronger collaboration between the New Taipei City High Riverbank Construction Management Office and the Department of Transportation is recommended. Improvement efforts should focus on the following two aspects:

(1) Infrastructure Improvement (*Responsible Agency: High Riverbank Construction Management Office*)

Priority should be given to key nodes identified as high-risk and high-detour locations, such as the boundary area between Sanchong and Banqiao Districts and other underserved peripheral areas. Improvements may include the construction of at-grade access ramps, bicycle ramps, and upgraded levee-crossing facilities. These measures would improve entrance efficiency and enhance the continuity of the bicycle network.

(2) Supporting Measures and Transfer Optimization (*Responsible Agencies: Department of Transportation and Local District Offices*)

For areas with relatively weak connections to transportation hubs, more comprehensive alternative routes and wayfinding systems should be established along urban streets and connecting roads. These improvements can reduce travel distances and provide clearer navigation for cyclists. For example, highly visible colored pavement markings, such as green or blue cycling lanes, could be installed along existing roads connecting MRT and railway stations (e.g., Yingge Station and Huilong Station) to riverside bicycle path entrances. In addition, continued efforts should be made to strengthen the integration between the YouBike system and the bicycle path network, thereby expanding the coverage of the green transportation system.

In conclusion, the Dahan River left-bank bicycle path has achieved a reasonable level of integration with the surrounding transportation system. Future improvements should focus on enhancing connections in underserved areas, upgrading critical entrance facilities, and further integrating bicycle-sharing services with public transportation. These measures would help eliminate network bottlenecks and significantly improve the continuity, transfer efficiency, and overall usability of the bicycle path network.

6. References

Newell, G. F. (1980). Traffic flow on transportation networks. MIT Press.

O'Sullivan, S., & Morrall, J. (1996). Walking distance to transit. *Transportation Research Record*, 1538(1), 19–26. <https://doi.org/10.1177/0361198196153800103>

Chou, K. Y., Paulsen, M., Nielsen, O. A., & Jensen, A. F. (2023). Analysis of cycling accessibility using detour ratios – A large-scale study based on crowdsourced GPS data. *Transportation Research Part D: Transport and Environment*, 117, 103669. <https://doi.org/10.1016/j.trd.2023.103669>

Martens, K. (2004). The bicycle as a feeding mode: experiences from three European countries. *Transportation Research Part D: Transport and Environment*, 9(4), 281–294.